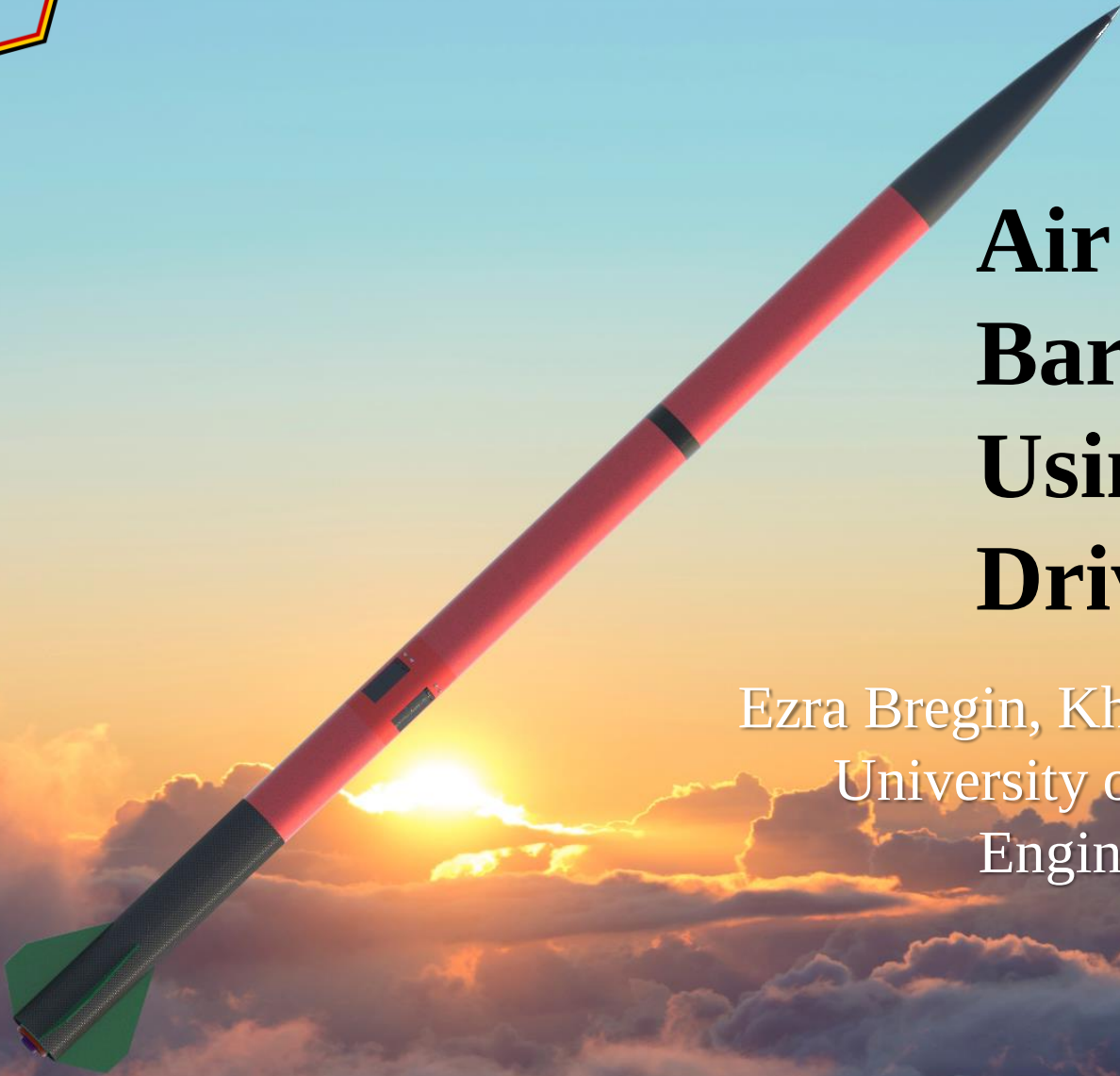




Terrapin Rocket Team

Air Brake Induced Barometric Error Correction Using a Wind Tunnel Data- Driven Approach

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Outline



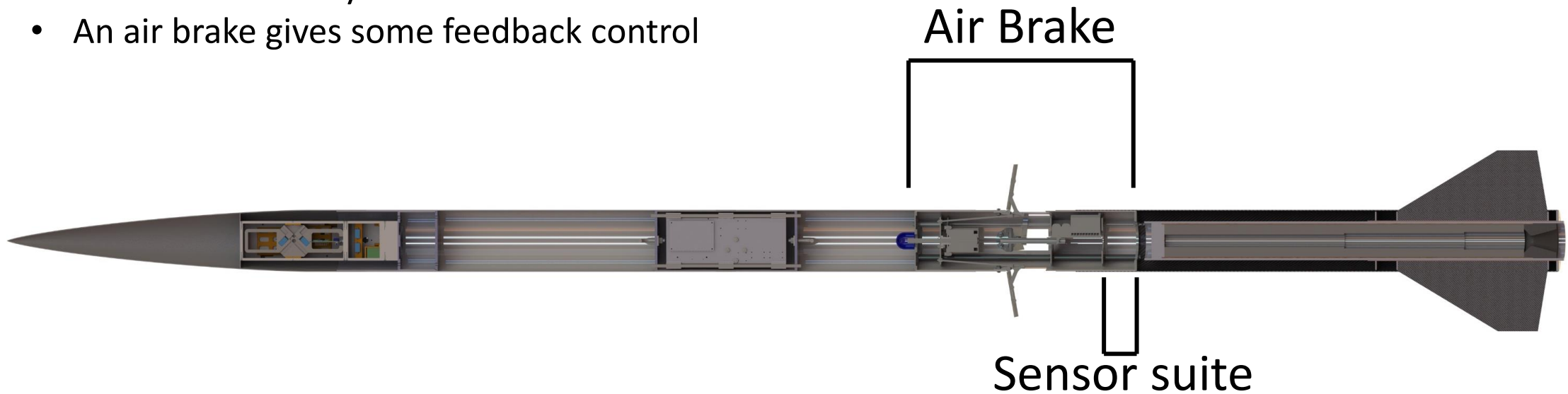
- Motivation
- Flap deployment induced pressure changes
- Data driven barometric error correction
- Low pass filter
- Results
- Conclusions
- Future work

Motivation

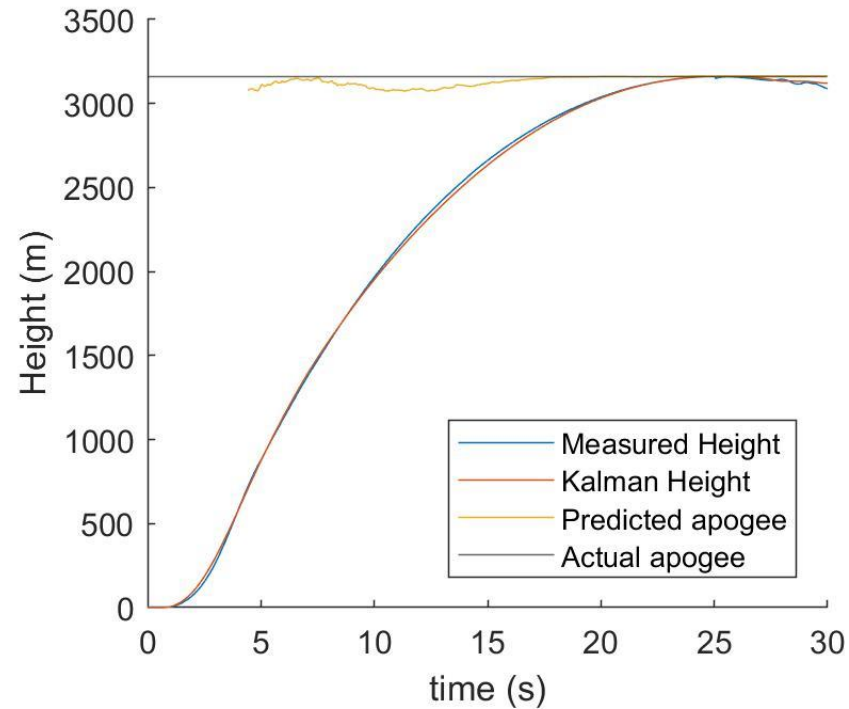


- Largest score opportunity at SAC 2023 is the apogee error
- Karkinos has a solid rocket motor
- Many uncertainties are not able to be modeled such as wind gusts
- Increase variability from SRAD motor
- An air brake gives some feedback control

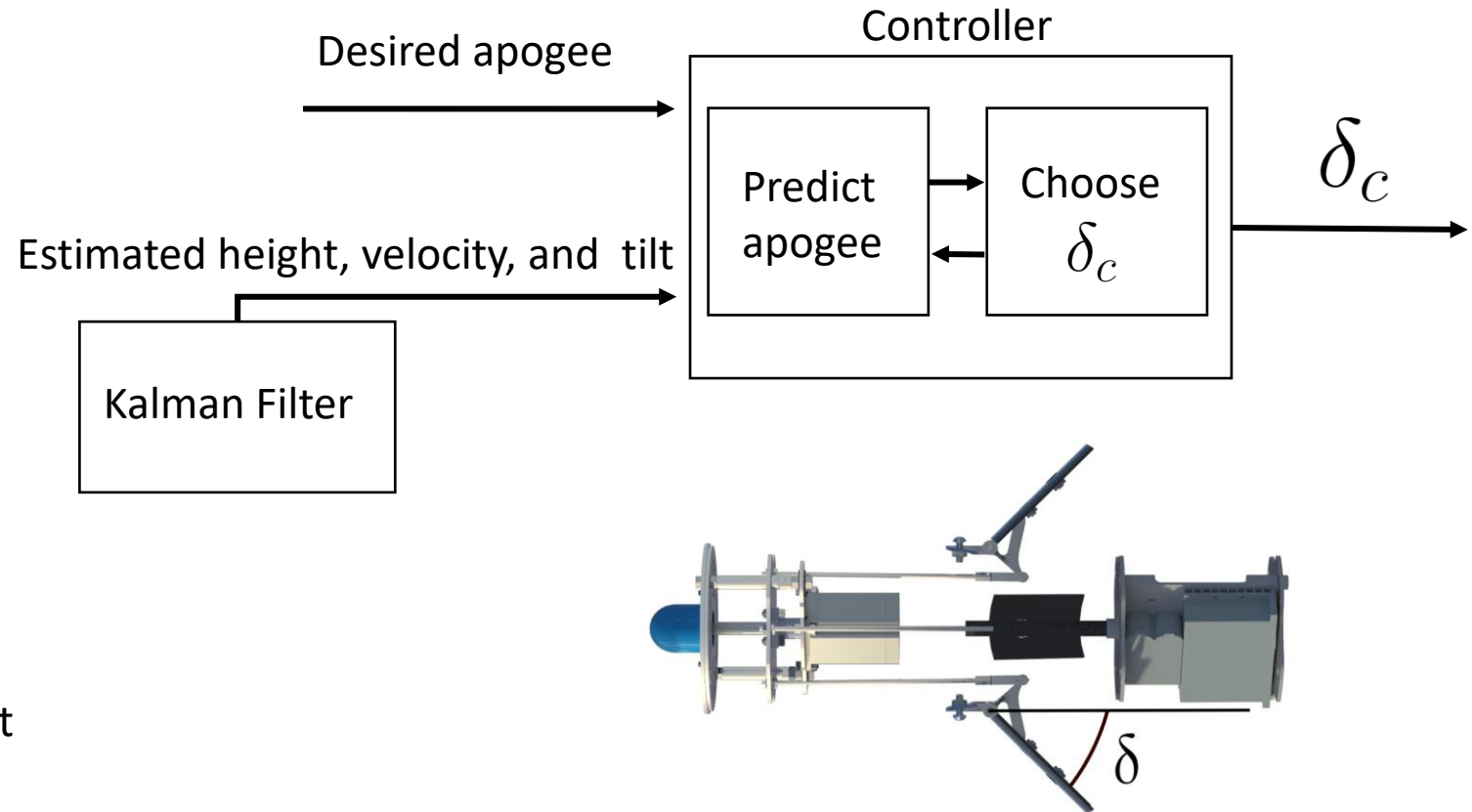
$$Points = 350 - \left(\frac{350}{0.3 \times Apogee_{Target}} \right) \times |Apogee_{Target} - Apogee_{Actual}|$$



Motivation



Height and predicted apogee without flap deployment

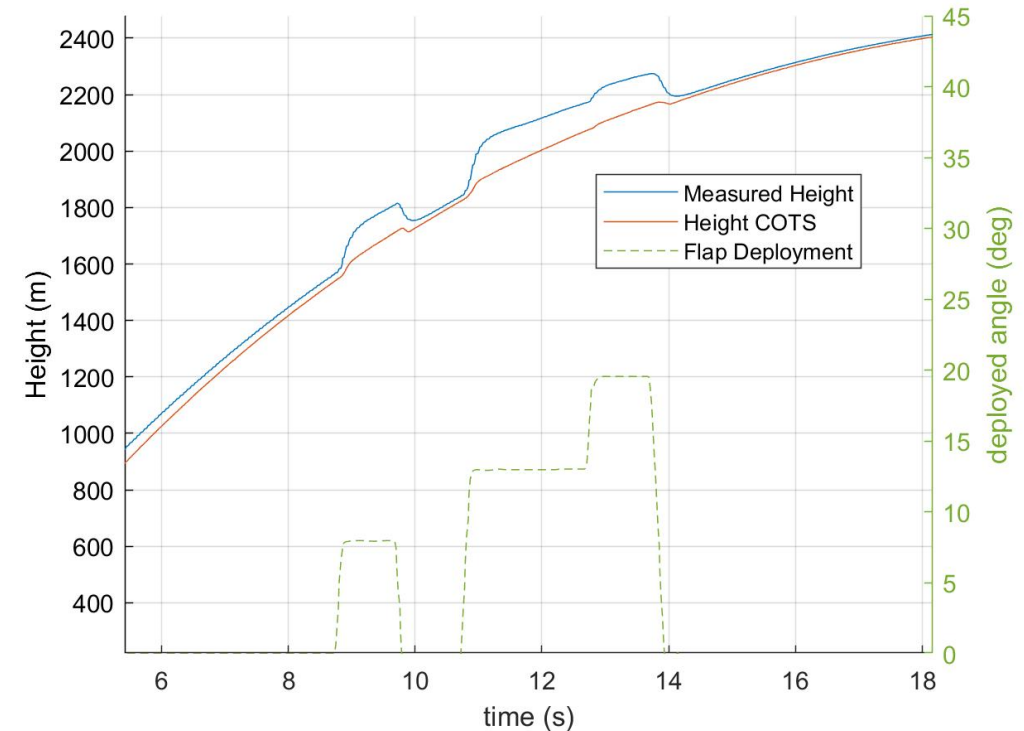


Accurate state estimation is extremely impactful on controller performance

Flap Deployment Induced Pressure Changes



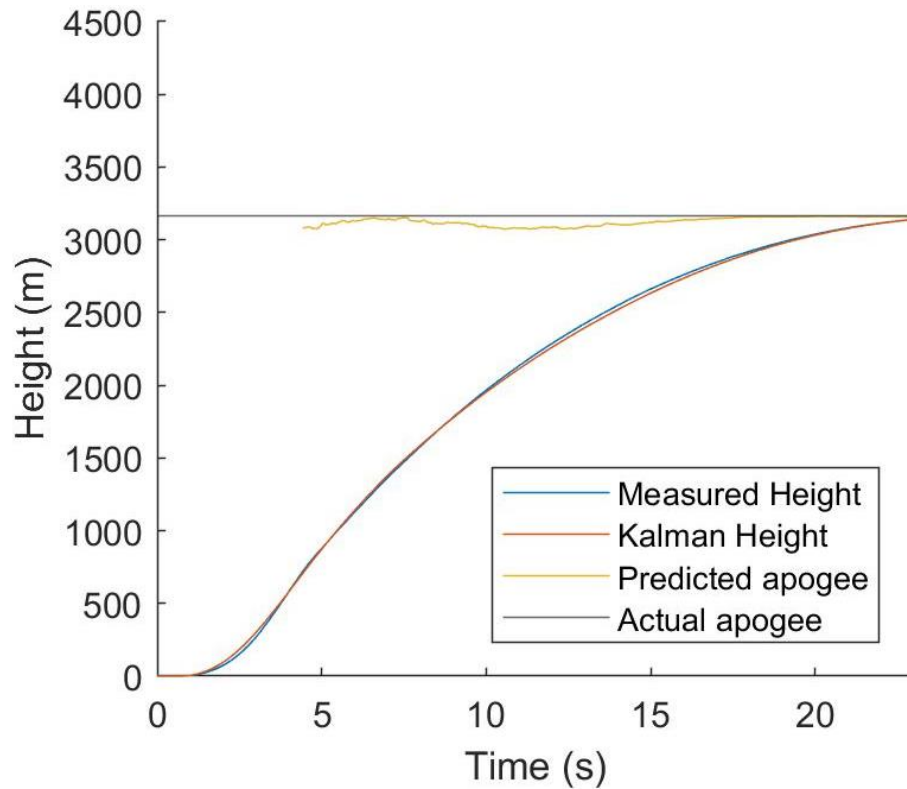
CFD simulation of Karkinos with varying flap deployment



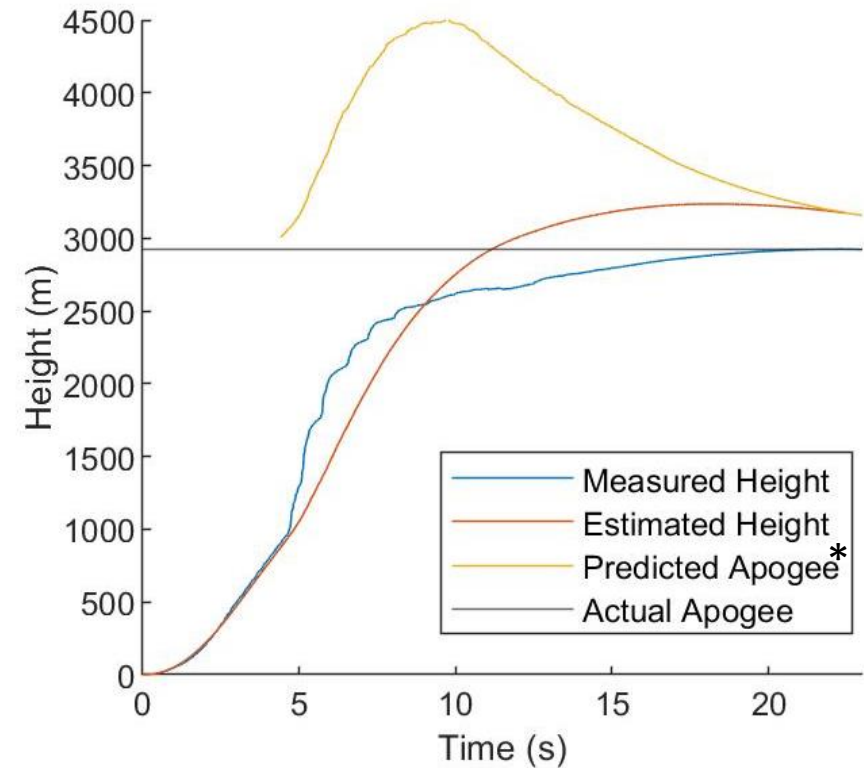
Measured height during flap deployment

Flap deployment induces height measurement error by reducing the local dynamic pressure

Flap Deployment Induced Pressure Changes



Karkinos 3/5/2023 Dormant Flaps



Karkinos 4/2/2023 Active Flaps

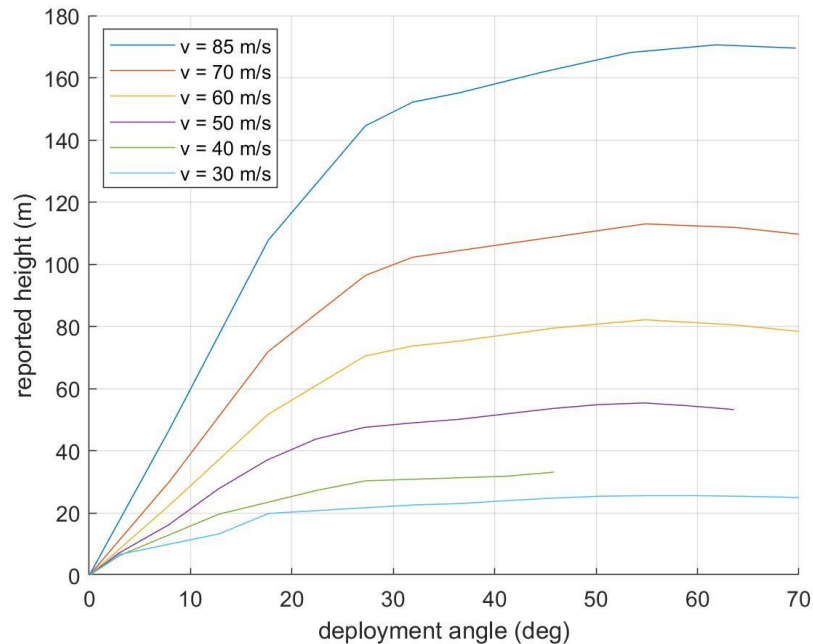
Control design is unfeasible while ignoring height measurement error

*Predicted apogee
assuming no flap
deployment

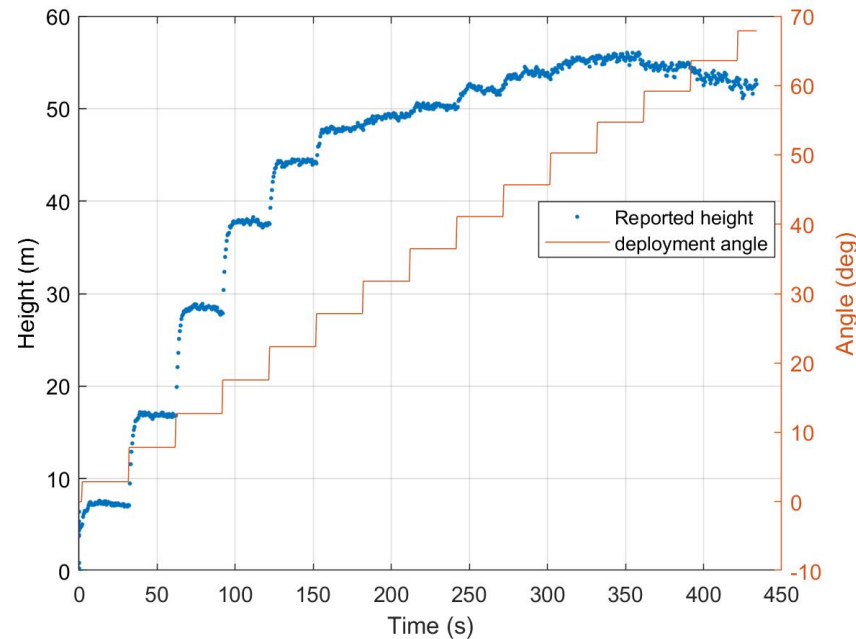
Data Driven Error Correction



- Need a quick minimal viable solution to get a working controller before SAC 2023
- Barometer height measurements were captured during wind tunnel tests



All wind tunnel tests



50 m/s test



Karkinos in the Glenn L. Martin Wind Tunnel

Data Driven Error Correction



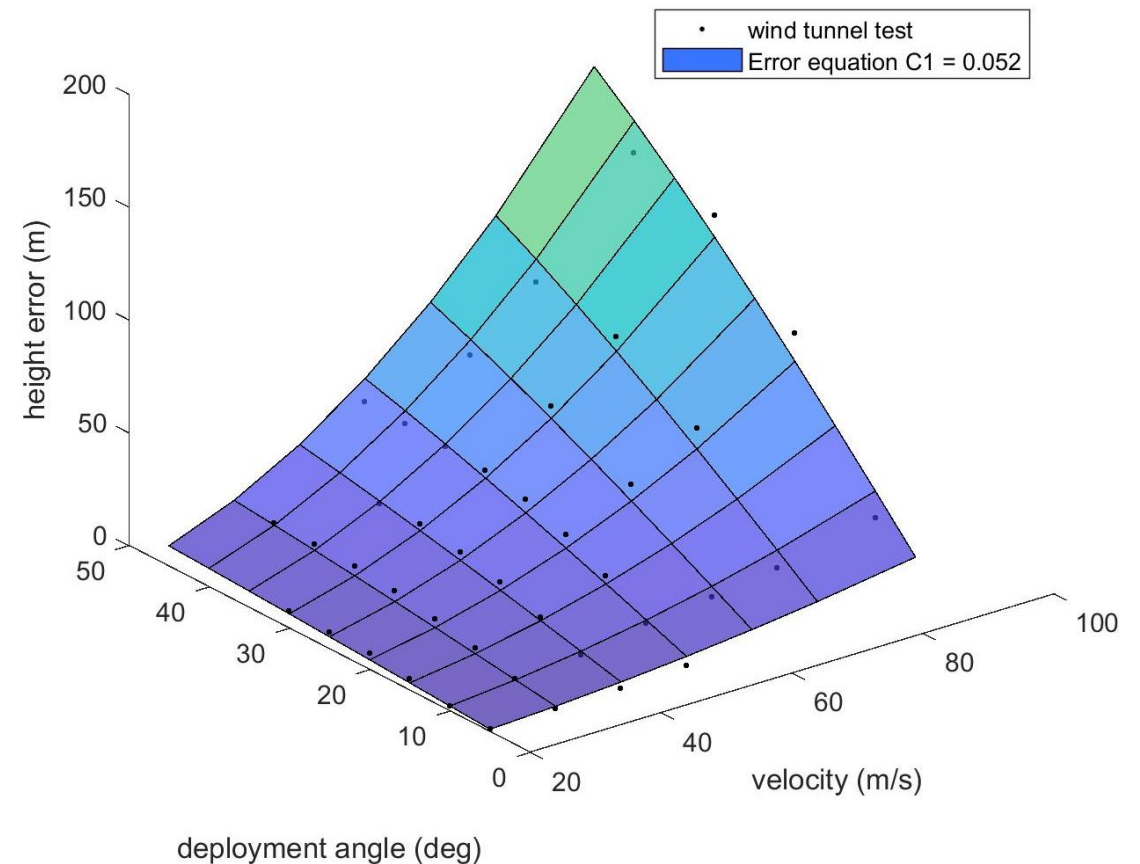
Upon careful inspection of the data and trying out a few error models, the following error model gave us the best function fit without overfitting,

$$h_r = h + f(h, v, \delta) \quad \text{Function of height, velocity and flap angle}$$

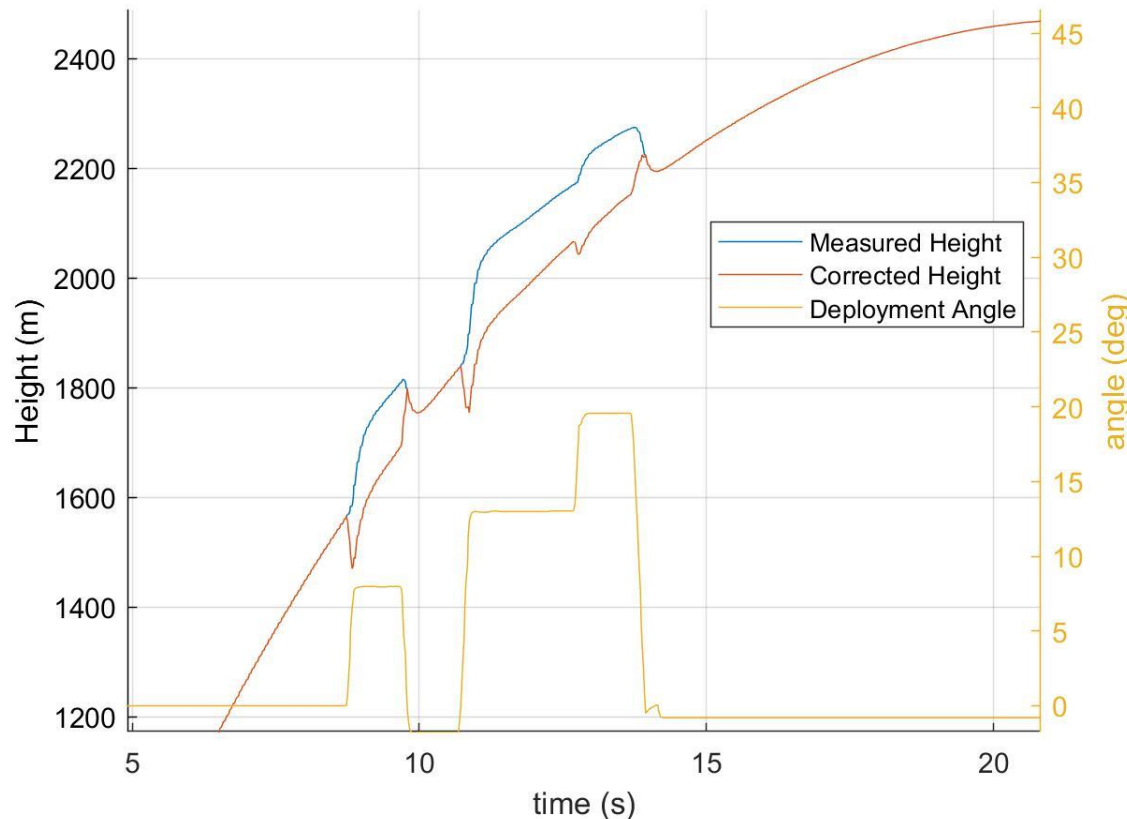
$$f(h, v, \delta) = \begin{cases} cq \sin \delta & \delta < 45^\circ \\ cq \sin 45^\circ & \delta \geq 45^\circ \end{cases}$$

q = dynamic pressure

- Linear least square fit to determine c
- $c = 0.052$ with $R^2 = 0.97$



Low-Pass filter



Karkinos 2/4/2023 deployment test

$$h_r = h + f(h, v, \delta)$$

$$f(h, v, \delta) = \begin{cases} cq \sin \delta & \delta < 45^\circ \\ cq \sin 45^\circ & \delta \geq 45^\circ \end{cases}$$

This error correction works well after steady-state
Does not work well during flap transitions

Can't use typical low-pass filter since its dependent
on the underling aerodynamics

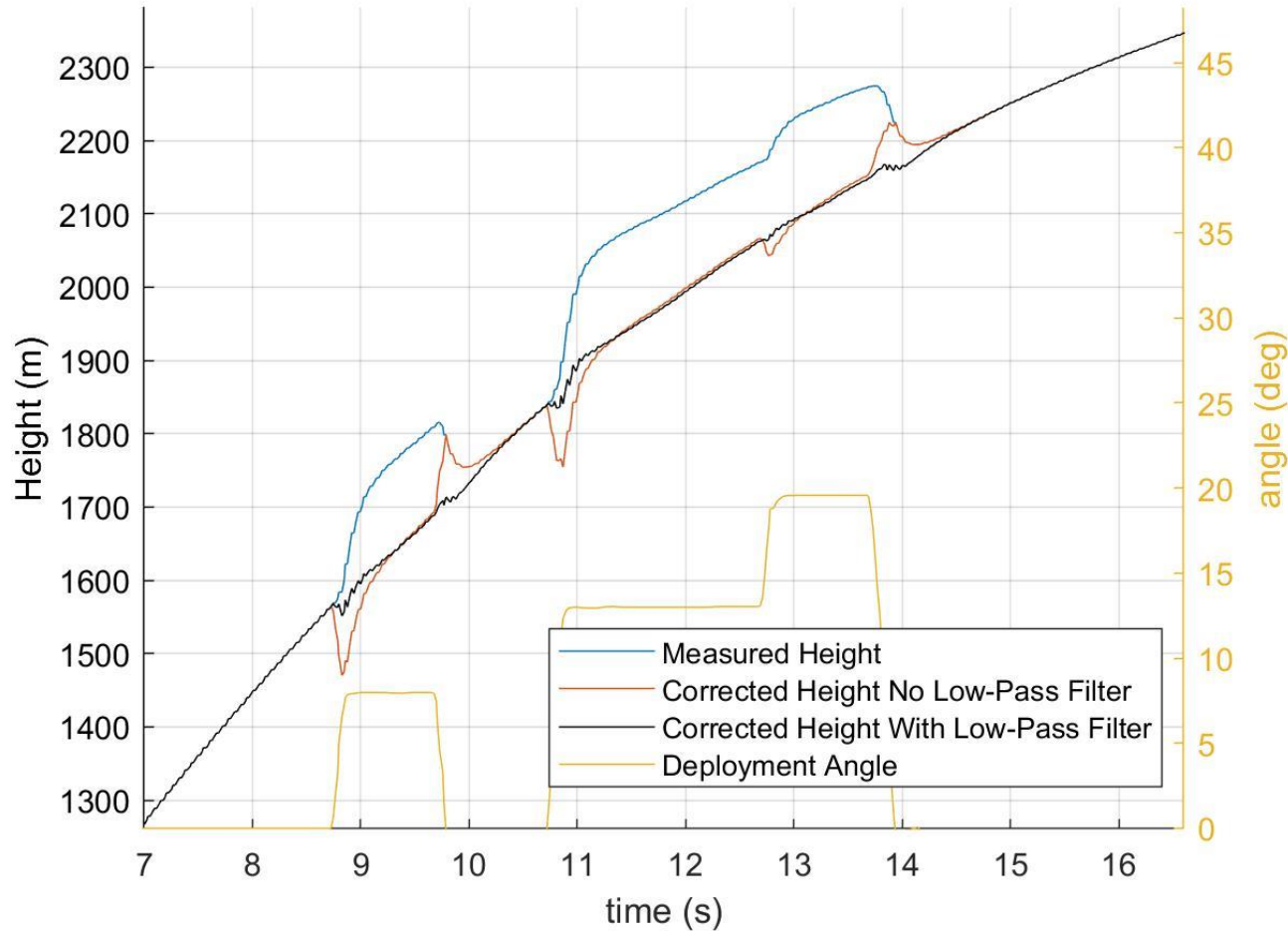
Use time dependent low-pass filter

$$\Delta_{k+1} = (1 - \alpha)\Delta_k + \alpha f(h, v, \delta)_k \quad \text{Error correction term}$$

$$\alpha \equiv \frac{T_s}{T_s + \tau}$$

T_s = sampling time

Low-Pass filter



$$\Delta_{k+1} = (1 - \alpha)\Delta_k + \alpha f(h, v, \delta)_k$$

$$\alpha \equiv \frac{T_s}{T_s + \tau}$$

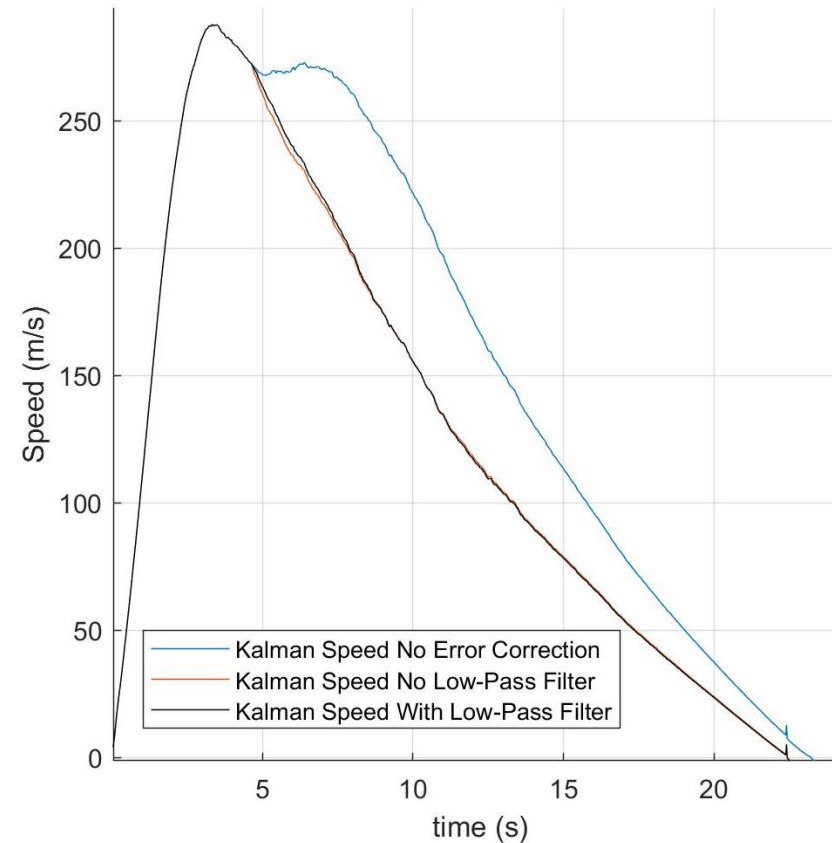
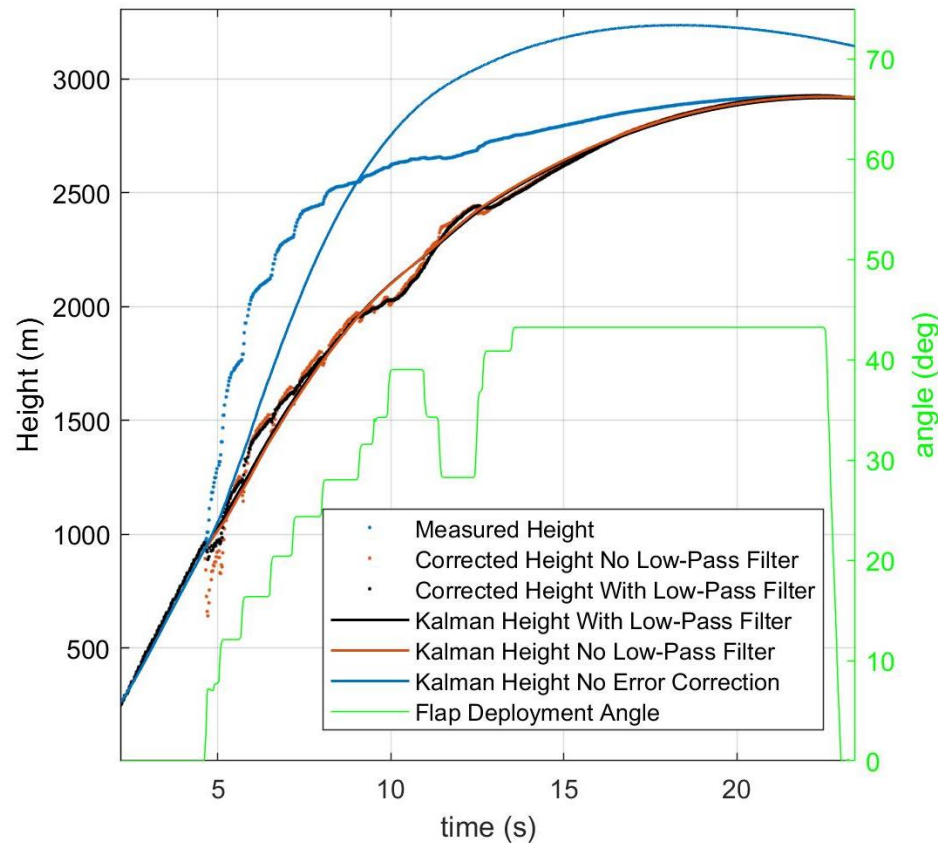
T_s = sampling time

τ was tuned using a Karkinos test flight

$$\tau = 0.15$$

Feed in this corrected height to the Kalman filter

Results Karkinos 4/2/23

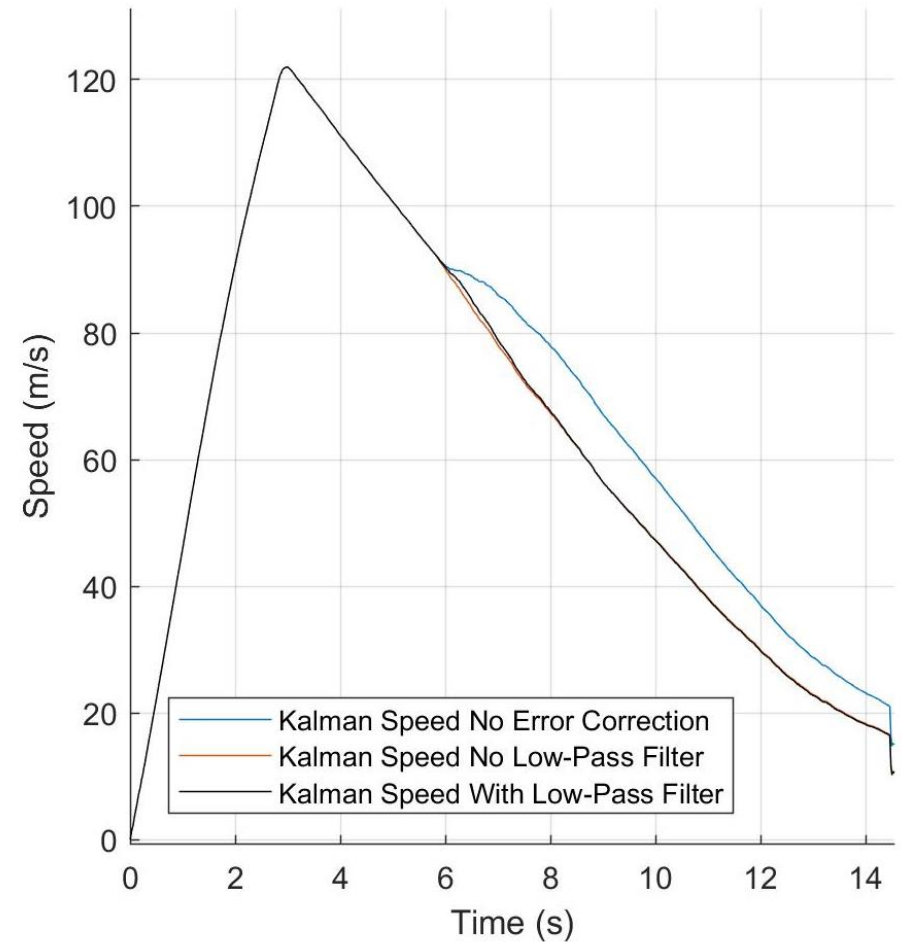
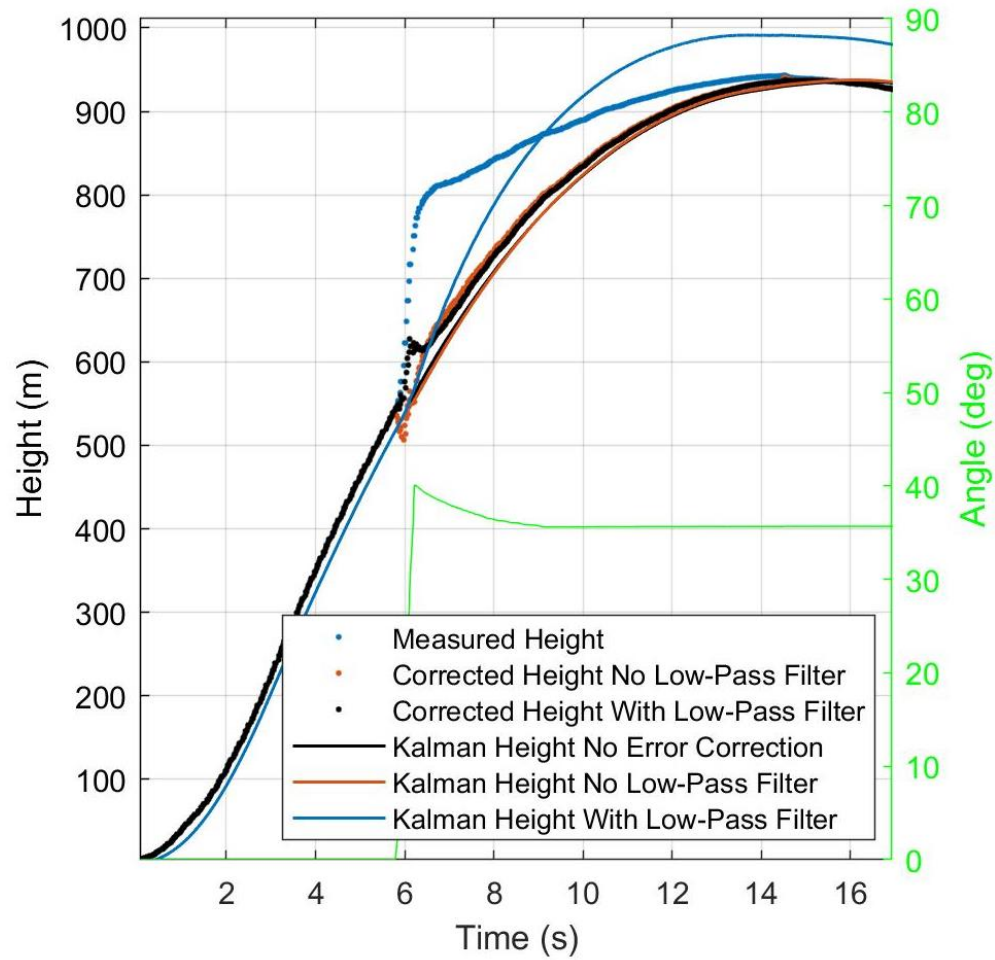


This error correction methodology works to estimate height and velocity

Results Icarus 5/6/23



Note that this flight was only to 900 meters



Conclusion



- Our heuristic data-driven error correction model works
- This error correction flew on Karkinos, and the air brake achieved a 74 ft apogee error
- The low pass filter method seems to work better but the Kalman filter can mostly handle the transient period

$$h_r = h + f(h, v, \delta)$$

$$f(h, v, \delta) = \begin{cases} cq \sin \delta & \delta < 45^\circ \\ cq \sin 45^\circ & \delta \geq 45^\circ \end{cases}$$



Flaps
deployed on
Icarus
5/6/2023

Future Work



- Retune the error correction parameters with the two new flights
- Look into other error models
- Look into error correction using the raw barometric measurement rather than the derived height output
- See if CFD analysis can properly characterize the change in dynamic pressure and create a height error look-up table

Questions?

